Curriculum Vitae
Toktam Rashidi Shahri (she/her)

Department of Physics & Astronomy, McMaster University, Hamilton, ON, Canada
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Research Interests

Astrophysical masers and 6.7 GHz methanol flares; superradiance and quantum coherent phenomena in astrophysics; radio flares and transients; numerical modelling of light–matter interaction; optimization and machine learning.

Education

Ph.D. Astrophysics Sep. 2023 – Present (expected defence: Summer 2027)
McMaster University, Canada *Hamilton, ON, Canada*
Supervisor: Dr. Fereshteh Rajabi; GPA: 3.84/4.0

M.Sc. Physics 2020 – 2023
University of Tehran, Iran *Tehran, Iran*
Advisor: Dr. Hossein Mohseni Sadjadi; GPA: 3.88/4.0

B.Sc. Physics 2016 – 2020
University of Tehran, Iran *Tehran, Iran*
GPA: 3.82/4.0

Publications and Manuscripts

For the most recent list, please see my Google Scholar profile.

Peer-Reviewed Publications

1. Fang, N., **Rashidi, T.**, Yonekura, Y., Anari, V., Burns, R., Houde, M., & Rajabi, F. (2026). *Modelling the 6.7 GHz Methanol Flaring in G358.93−0.03 Using the Maxwell–Bloch Framework.* New Astronomy, in press (arXiv:2610.20506).
2. **Rashidi, T.**, Anari, V., Powles, O., MacLeod, G. C., Tanabe, Y., Yonekura, Y., & Rajabi, F. (2026). *A Unified Maxwell–Bloch Framework for Multiperiodic 6.7 GHz Methanol Flaring in G9.62+0.20E.* Monthly Notices of the Royal Astronomical Society, 545(3), staf2200 (arXiv:2510.16265).
3. **Rashidi, T.**, Anari, V., Bartkiewicz, A., Wolak, P., Szymczak, M., & Rajabi, F. (2025). *Modelling the Periodic 6.7 GHz Methanol Flaring in G22.356+0.066.* Monthly Notices of the Royal Astronomical Society: Letters, 542(1), L12–L18 (arXiv:2409.09209).

Manuscripts Submitted and under Review

1. Anari, V., **Rashidi, T.**, & Rajabi, F. (2026). *Fast Population Leakage in Astronomical Masers: Maser Amplification and Transient Superradiance.* Under review in New Astronomy (arXiv:2608.00332).
2. Fang, N., **Rashidi, T.**, MacLeod, G. C., Houde, M., & Rajabi, F. *Modelling the Multitransition Periodic Flaring in G323.46−0.08 Using the Maxwell–Bloch Framework.* Submitted to New Astronomy, November 2026.

Manuscripts in Preparation

1. **Rashidi, T.**, Liu, Z., Anari, V., Yonekura, Y., Bartkiewicz, A., Wolak, P., & Rajabi, F.
Maxwell–Bloch Modelling of Periodic 6.7 GHz Methanol Flares. I. Stable Quiescent Emission.
2. **Rashidi, T.**, Liu, Z., Anari, V., Yonekura, Y., MacLeod, G. C., & Rajabi, F.
Maxwell–Bloch Modelling of Periodic Flare Profiles in the 6.7 GHz Methanol Transition. II. Evolving Quiescent Baselines and Complex Flare Structure.
3. Liu, Z., **Rashidi, T.**, Anari, V., Yonekura, Y., & Rajabi, F.
From Maser Amplification to Superradiance: Maxwell–Bloch Modelling of Methanol Flares in G14.230−0.509.

Research Experience

Research Intern Sep. 2026
Piwnice Astronomical Observatory, Nicolaus Copernicus University *Toruń, Poland*
Host: Dr. Anna Bartkiewicz

- Learned the 6.7 GHz methanol maser monitoring and data-reduction methods behind the Toruń light curves, and presented the Maxwell–Bloch modelling to the observing group.

Research Assistant Sep. 2023 – Present
Department of Physics & Astronomy, McMaster University *Hamilton, ON, Canada*
Supervisor: Dr. Fereshteh Rajabi

- Contributed to the optimization and enhancement of an existing one-dimensional Maxwell–Bloch solver, including improvements in computational efficiency and usability, as well as to the development of the three-level Maxwell–Bloch solver.
- Developed and applied period-determination and flare-morphology diagnostics to several flaring 6.7 GHz methanol sources, contributed to the coherence-diagnostics analysis, and performed Maxwell–Bloch modelling of the observed flares.

Research Assistant Sep. 2020 – Feb. 2023
University of Tehran *Tehran, Iran*
Advisor: Dr. Hossein Mohseni Sadjadi

- Investigated Gauss–Bonnet and mass-varying neutrino couplings and their impact on late-time cosmic expansion and neutrino-mass evolution.

Presentations

Talks

1. **(Invited) Modelling Variability in Maser Light Curves**. Piwnice Astronomical Observatory, Toruń, Poland. 22 September 2026.
2. **A Race to Coherence: Superradiance and Competing Channels in a Λ-System**. 9th Annual Toronto Ultracold Atom Network (TUCAN), University of Toronto, Toronto, ON, Canada. 13 May 2026.
3. **Cooperative Spontaneous Processes and Astronomical Flares**. McMaster Theory Journal Club, McMaster University, Hamilton, ON, Canada. 1 April 2026.
4. **The Effect of Fast Parallel Transitions on a Superradiance Cascade**. McMaster Astronomy Journal Club, McMaster University, Hamilton, ON, Canada. 11 December 2025.
5. **Dicke's Superradiance and Astrophysical Masers: Modelling Flares in G22.356+0.066 and G9.62+0.20E**. Ralposium, McMaster University, Hamilton, ON, Canada. 25–27 June 2025.

6. **Superradiance and Periodic Methanol Flaring in G22.356+0.066**. Physics and Astronomy Graduate Symposium, McMaster University, Hamilton, ON, Canada. 17 October 2024.
7. **Dicke's Superradiance and Periodic Flaring in G22.356+0.066**. Star Formation Workshop, McMaster University, Hamilton, ON, Canada. 12–14 August 2024.

Posters

1. **Unveiling Superradiant Signatures in Periodic Methanol Flares**. CASCA, Montreal, QC, Canada. 9–11 June 2026.
2. **A Unified Maxwell–Bloch Framework to Model Flaring Behaviour in Maser-Hosting Regions**. The 3rd International Online Conference on Universe. 4–6 March 2026.
3. **Dicke's Superradiance in Astrophysical Masers: Modelling Periodic Flaring in G9.62+0.20E**. Women+ in Physics Canada 2025, Queen's University, Kingston, ON, Canada. 22–25 July 2025.
4. **Light-Matter Dynamics Behind Flares in Maser-Hosting Regions**. The 19th Women in Science and Engineering Annual National Research Conference, McMaster University, Hamilton, ON, Canada. 7 March 2025.

Funding & Awards

Physics and Astronomy Departmental Award ($5,000 (×2)) 2025 and 2026
McMaster University

School of Graduate Studies Grant in Aid for Research Travel and Field Work ($2,950) 2025–2026
McMaster University

Direct Admission to M.Sc. Program 2020
University of Tehran; awarded for outstanding undergraduate GPA

Tuition Waiver for M.Sc. Studies 2020–2023
University of Tehran

Tuition Waiver for B.Sc. Studies 2016–2020
University of Tehran; awarded for top performance in the Iranian National University Entrance Exam

Teaching & Mentorship

Research Mentorship

Mentored students in Dr. F. Rajabi's research group and collaborated with them on Maxwell–Bloch modelling and light-curve analysis.

- Zhejian Liu, master's student, McMaster (Jan. 2026 – present); co-author of three manuscripts, first author of one
- Oscar Powles, undergraduate, McMaster (Sep. 2024 – Jul. 2025); co-author of Rashidi et al. 2026, MNRAS
- Subhradeep Mahata, MITACS student, McMaster (summer 2024).

Teaching Assistantships

Teaching Assistant Sep. 2023 – Present
McMaster University *Hamilton, ON, Canada*
Scientific Computing; Introductory Mechanics; Quantum Mechanics II; Waves, Electricity, and Magnetism; Introduction to Modern Physics

Teaching Assistant Sep. 2020 – Dec. 2021
University of Tehran *Tehran, Iran*
Advanced Quantum Mechanics; Symmetries and Group Theory; Fundamentals of Elementary Particles; General Physics I

Course Consultant | Undergraduate Teaching Seminar, McMaster University Winter 2025 & Winter 2026
Advised a student group preparing teaching presentations on particle physics for ISCI 3A12: Light, the Universe, and Everything.

Service, Outreach & Community Engagement

Event Organizer Sep. 2025 – Present
Promoting Inclusivity in Physics & Astronomy (PIPA), McMaster University *Hamilton, ON, Canada*
Organize and coordinate student-led events, including coffee hours and movie nights.

Presenter Sep. 2025 – Present
McMaster W. J. McCallion Planetarium *Hamilton, ON, Canada*
Deliver planetarium presentations and support astronomy outreach activities.

Judge 16–17 Oct. 2026
Canadian Undergraduate Physics Conference (CUPC), McMaster University *Hamilton, ON, Canada*
Evaluated oral presentations.

Chair and Judge 15 Oct. 2026
Physics and Astronomy Graduate Symposium Day, McMaster University *Hamilton, ON, Canada*
Chaired the astronomy session and evaluated the talks.

Astronomy Outreach Facilitator Winter 2025 & Winter 2026
McMaster University *Hamilton, ON, Canada*
Co-hosted kindergarten groups visiting the Planetarium and helped design age-appropriate space-themed craft and learning activities.

Local Organizing Committee Spring 2024
Toronto Ultracold Atom Network, McMaster University *Hamilton, ON, Canada*
Supported preparation and logistics.

Course Assistant and Event Organizer 2017 – 2020
University of Tehran *Tehran, Iran*
Helped move General Physics to an online format and organized departmental and student events.

Organizer and Facilitator Winter 2019 – Spring 2020
Weekly Feminist Reading Circle, University of Tehran *Tehran, Iran*
Initiated and coordinated weekly reading sessions and selected reading materials.

Skills & other Details

Programming languages: Python; C++; MAPLE
Research methods: Numerical analysis; data analysis; mathematical modelling
Languages: Persian: native; English: fluent; French: elementary